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0178/04

October/November 2016

2 hours 30 minutes

Candidate's answer on the Question Paper.

Additional Materials: Electronic Calculator
Geometrical Instruments
Tracing Paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in blue or black ink.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer all questions.

If working is needed for any question it must be shown below that question.
Electronic calculators should be used.

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If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π use either your calculator value or 3.142.

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The number of marks is given in brackets [] at the end of each question or part question.
The total of the marks for this paper is 120.

The total of the marks for this paper is 130.

- 1 Lipuo makes a sequence of patterns using black and grey triangular tiles.



pattern
number
1



pattern
number
2



pattern
number
3

pattern
number
4

- (a) Draw the fourth pattern in the space provided.

[1]

- (b) Complete the table.

Pattern	Number of black tiles	Number of grey tiles
1	1	3
2	1	6
3	1	9
5	1	
6	1	
$n + 2$	1	

[3]

- (c) Find the number of grey tiles in the 15th pattern.

Answer (c) [1]

- (d) Find the pattern number that can be made from 61 tiles.

Answer (d) [2]

- (e) Find, in terms of n , the **total** number of tiles in the n th pattern.

Answer (e) [2]

- 2 Mr Tau invests M10 000 at $r\%$ **simple interest** per year.
He receives a total of M11 500 at the end of 3 years.

(a) Calculate the value of r .

Answer (a) $r = \dots\dots\dots$ [3]

- (b) Mr Tau then uses 20% of the M11 500 to pay school fees.

Calculate the amount paid for the school fees.

Answer (b) M $\dots\dots\dots$ [2]

- (c) He also shares M5000 among his daughters Lineo and 'Mapule in the ratio 2 : 3.

(i) Calculate the amount Lineo receives.

Answer (c)(i) M $\dots\dots\dots$ [2]

- (ii) 'Mapule takes M1000 from her share and invests it at 10% **compound interest** per year.

After how many years will her investment be worth M1331?

Answer (c)(ii) $\dots\dots\dots$ years [3]

- (d) 'Mapule sells a watch for M180 making a profit of 20%.

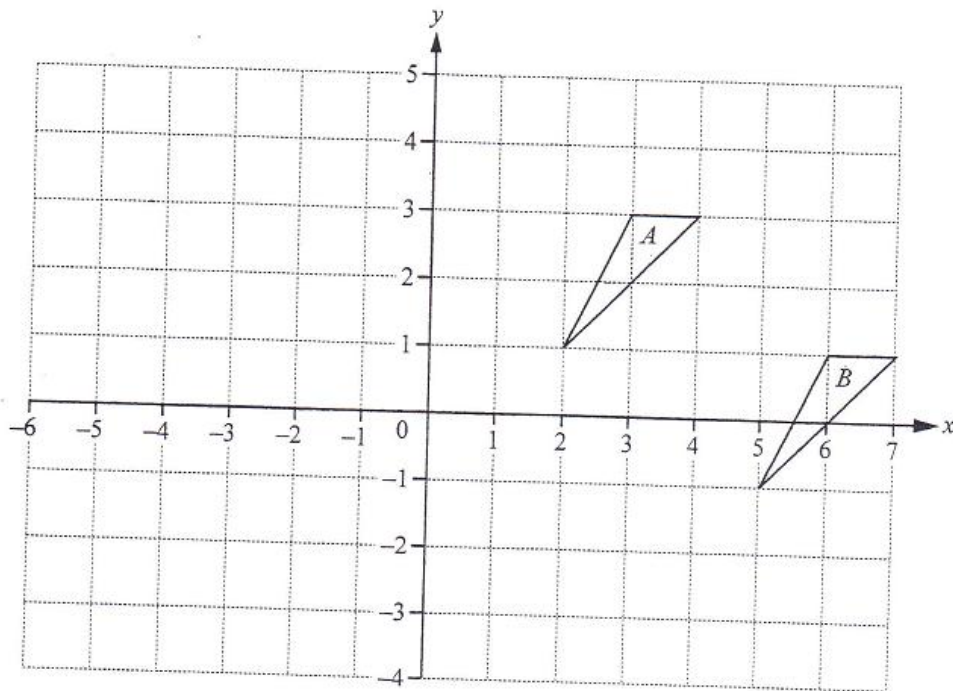
Calculate the price 'Mapule paid for the watch.

Answer (d) M $\dots\dots\dots$ [2]

- (e) Lineo bought a watch costing M240.
She later sold it for M196.

Calculate her percentage loss.

- 3 The diagram shows triangle A and triangle B .



- (a) Describe fully the single transformation which maps triangle A onto triangle B .

Answer (a) [2]

- (b) Draw the enlargement of triangle A , centre $(1, 1)$, scale factor -2 and label the image C . [2]

- (c) Reflect triangle A in the line $y = x$ and label the image D . [2]

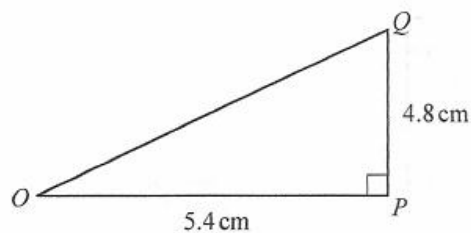
- (d) A transformation is represented by the matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

- (i) Draw the image of triangle A under this transformation. Label the image E . [2]

- (ii) Describe fully the single transformation which maps triangle A onto triangle E .

Answer (d)(ii) [3]

- 4 (a) The diagram shows triangle OPQ .
Angle $OPQ = 90^\circ$, $OP = 5.4$ cm and $PQ = 4.8$ cm.



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- (i) Calculate OQ .

Answer (a)(i) cm [2]

- (ii) Calculate the gradient of line OQ .
Give your answer correct to 2 significant figures.

Answer (a)(ii) [3]

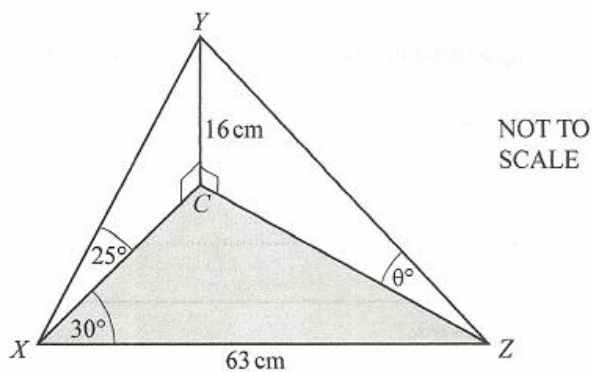
- (iii) OP and PQ are measured correct to 2 significant figures.

Calculate the greatest possible area of triangle OPQ .

Answer (a)(iii) cm^2 [3]

- (b) In the diagram, XZ has length 63 cm and YC is a vertical line of length 16 cm.
Points X , Z and C lie on a horizontal plane.

The angles of elevation of Y from X and Z are 25° and θ° respectively.
Angle $CXZ = 30^\circ$.



- (i) Calculate the length of XC .
Give your answer correct to 4 significant figures.

Answer (b)(i) cm [2]

- (ii) Show that the length of $ZC = 37.4$ cm correct to 3 significant figures.

[4]

- (iii) Find the value of θ .

Answer (b)(iii) [2]

- (iv) Calculate the area of triangle XZC .

Answer (b)(iv) cm^2 [3]

- (v) Find the shortest distance from C to XZ .

Answer (b)(v) cm [2]

- 5 The table shows the time, y years, that 150 men have been married for.

Time (y years)	$0 < y \leq 10$	$10 < y \leq 20$	$20 < y \leq 30$	$30 < y \leq 40$	$40 < y \leq 50$
Frequency	30	50	35	25	10

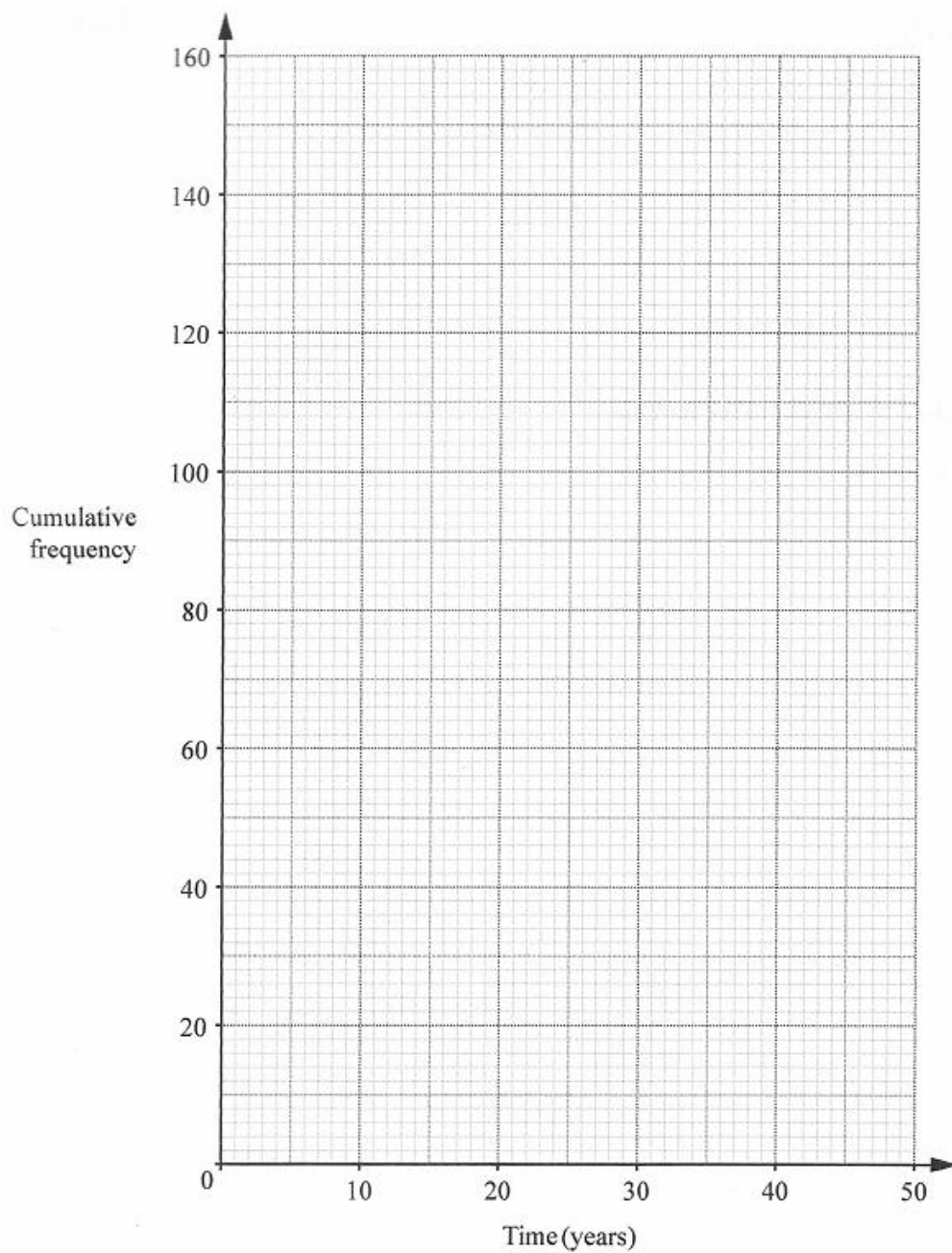
- (a) Calculate an estimate of the mean.

Answer (a) years [4]

- (b) The table shows the cumulative frequencies.

Time (y years)	$y \leq 10$	$y \leq 20$	$y \leq 30$	$y \leq 40$	$y \leq 50$
Cumulative frequency	30	80	115	140	150

- (i) Use the information from the table to draw a cumulative frequency curve on the grid.



[3]

- (ii) Use your graph to find

- (a) the interquartile range,

Answer (b)(ii)(a) years [2]

- (b) the number of men who have been married for more than 26 years.

Answer (b)(ii)(b) [2]

- 6 (a) Simplify.

$$(4x^2y^3)^4$$

Answer (a) [2]

- (b) Make a the subject of the formula.

$$P = \frac{r}{\sqrt{a}}$$

Answer (b) $a =$ [2]

(c) Simplify fully.

$$\frac{2x^2 - 5x - 3}{x^2 - 3x}$$

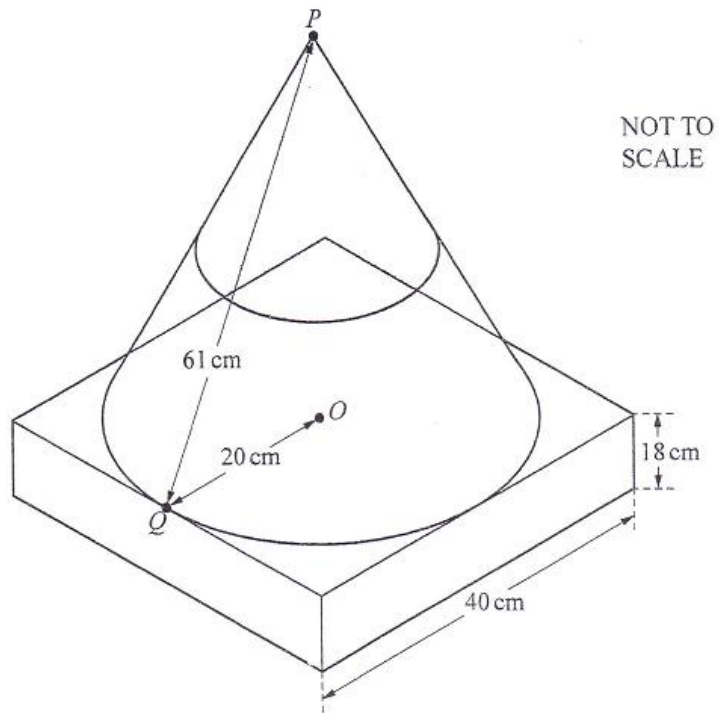
Answer (c) [3]

(d) Solve $x^2 + 5x - 3 = 0$.

Answer (d) $x =$ or $x =$ [4]

- 7 A “traffic cone” is made from a **cone** and a **cuboid**.

The cone has radius $OQ = 20$ cm and slant height $QP = 61$ cm.
The cuboid has a square base, of side 40 cm and height 18 cm.



- (a) Sketch the net of the conical part of the “traffic cone”.

- (b) How many planes of symmetry has the “traffic cone”?

Answer (b) [1]

- (c) Calculate the **total** surface area of the “traffic cone”.
Give your answer correct to the nearest hundred cm^2 .

[The curved surface area, A , of a cone with radius r and slant height l is $A = \pi rl$.]

Answer (c) cm^2 [7]

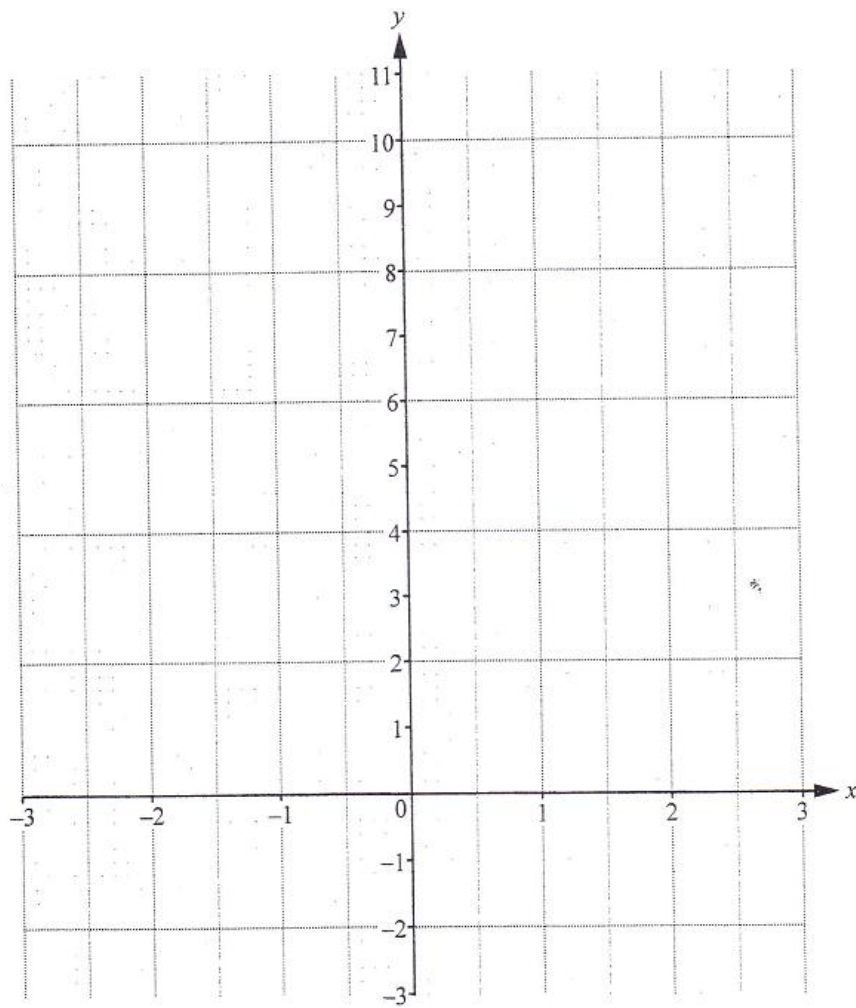
- 8 The table shows values of the function $y = \frac{1}{3}x^3 - x + 4$.

x	-3	-2.5	-2	-1.5	-1	0	1	1.5	2	2.5	3
y	-2		3.3	4.4	4.7	4	3.3			6.7	10

(a) Complete the table.

[3]

(b) On the grid, draw the graph of $y = \frac{1}{3}x^3 - x + 4$ for $-3 \leq x \leq 3$.



[4]

- (c) Use your graph to solve the equation $\frac{1}{3}x^3 - x = 0.2$.

Answer (c) $x = \dots\dots\dots$ or $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [3]

- (d) By drawing a suitable straight line on the same grid, solve the equation $\frac{1}{3}x^3 - x + 4 = x + 4$.

Answer (d) $x = \dots\dots\dots$ or $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

- (e) The equation $\frac{1}{3}x^3 - x + 4 = x + 4$ can be written in the form $Ax^3 + Bx = 0$ where A and B are integers.

Find the values of A and B .

Answer (e) $A = \dots\dots\dots$ and $B = \dots\dots\dots$ [3]

- 9 The functions, f and g , are defined by

$$f(x) = \frac{x}{3} + 1 \text{ and } g(x) = 2x - 1.$$

- (a) Calculate $g(-3)$.

Answer (a) [1]

- (b) Find, in its simplest form,

(i) $f^{-1}(x)$,

Answer (b)(i) [2]

(ii) $fg(x)$,

Answer (b)(ii) [2]

(iii) $(fg)^{-1}(x)$.

Answer (b)(iii) [2]

(c) Find $g^{-1}\left(2\frac{1}{2}\right)$.

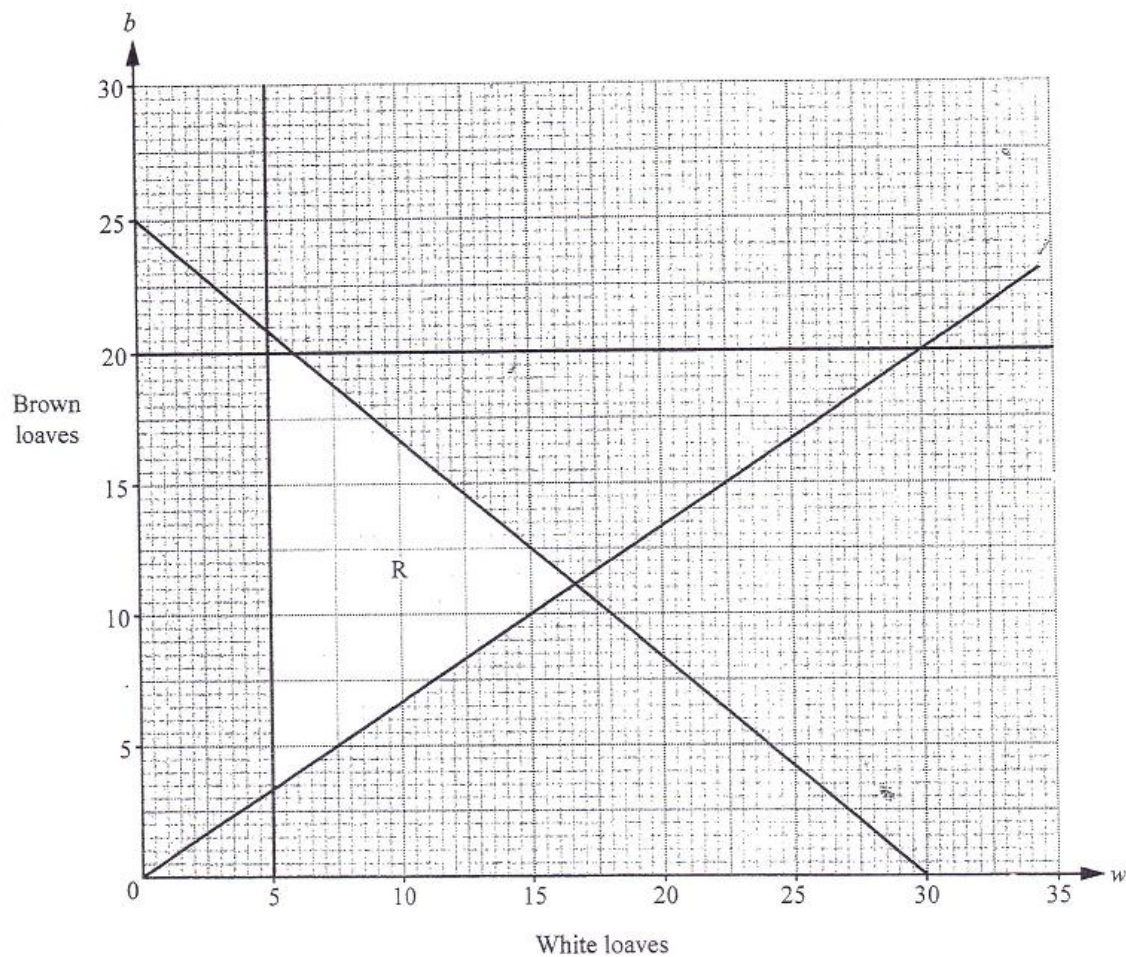
Answer (c) [3]

(d) Find the value of x for which $gf(x) = 5$.

Answer (d) $x =$ [4]

- 10 The region, R, in the grid represents the number of white (w) and brown (b) loaves of bread produced in a bakery.

One of the inequalities defining the region, R, is $b \leq 25 - \frac{5}{6}w$.



- (a) Write the other three inequalities defining region R.

Answer (a)

.....

..... [3]

- (b) If the profit made on a loaf of white bread is M1.80 and the profit made on a loaf of brown bread is M1.40, write an equation for the total profit, P .

Answer (b) $P = \dots\dots\dots$ [1]

- (c) Calculate the maximum profit that can be made on selling white and brown loaves of bread.

Answer (c) M $\dots\dots\dots$ [3]

- (d) In one month the profit on a loaf of white bread decreased to M1.60 and on a loaf of brown bread increased to M1.60.

- (i) Determine the new profit equation.

Answer (d)(i) $P = \dots\dots\dots$ [1]

- (ii) Calculate the profit made by selling 15 loaves of white bread and 10 loaves of brown bread.

Answer (d)(ii) M $\dots\dots\dots$ [2]

- (e) State if the new profit equation will affect the maximum profit.
Support your answer by showing all the calculations.